LOAD MAPS – THE WEB PAGE

CTRN 1:5000 maps for Garmin GPS from the civil protection Drive, with nothing to install · English translation: in case of doubt the Italian text prevails · © 2026 Claude René Monbaron – UCS PC ANA PN e PC RFVG

Address

https://claudemonbaron.github.io/Carica_Mappe/

Opens in any browser, on computer, tablet or phone. Worth bookmarking (on a phone: Share › Add to Home Screen).

What it does

1. Reads the up-to-date list of maps from the Drive: all Italian regions, Switzerland, Slovenia and Austria.
2. You choose the area; it downloads the maps and, if there are several municipalities, **joins them into a single file** ready for the GPS (GPSMAP, Atemos and other Garmin units that read .img maps).
3. Maps already downloaded stay in the browser: the second time the same area loads at once.

1 · Area

1. Choose **Region / Country** and **Province** (Land or Canton abroad), then type the **Municipality**.
2. **Province**: the map of the whole province. **Municipality**: only that municipality. **Neighbours**: the municipality and all those bordering it, also across the border. **Rectangle**: the municipality plus all municipalities within the given km towards North, South, East and West (5 km by default).
3. In the **Chosen maps** list a municipality can be removed with its tick box. Below: number of maps, MB and **tiles**.

2 · On a computer with Chrome or Edge: straight into the GPS

1. Connect the GPS with the cable (or the memory card with a reader).
2. **Choose the Garmin folder of the GPS**: select the `Garmin` folder of the GPS and allow Chrome to edit the files. The page remembers it; next time just **Reactivate access to the GPS**.
3. The list of maps already on the GPS appears with their tiles; those no longer needed can be **deleted** from here.
4. Choose **Internal memory** or **Memory card**, then **Load into GPS**. At the end a green «Fatto» (Done) appears.
5. **GPX tracks** (assigned areas, tracks to hand back): **Select GPX…** picks the files and adds them to the list; it can be repeated from other folders, and a file is removed with **Remove**. With the box **Also load to the GPS with the map**, pressing **Load to GPS** puts the GPX of the list into the `Garmin/GPX` folder of the GPS together with the map, replacing those with the same name. The list empties when the page is closed.
6. **Eject** the GPS (Finder or File Explorer) before unplugging the cable.
7. Without a GPS connected: **Download the file (without GPS)** prepares the same joined file in Downloads, to copy to the GPS later or from another computer.

3 · Phone, tablet, Safari or Firefox: with a file to copy

1. Once the area is chosen, press **Download the joined file** (or **Download the file** for a single map): the file goes to Downloads.
2. Connect the GPS and copy the file into its `Garmin` folder: on iPhone with the **Files** app, on Android with the file manager (OTG cable), on a computer with Finder or File Explorer.

3. For very large provincial maps (hundreds of MB), on a phone the link **«download the file directly from the Drive»** below the button is better.

Good to know

1. **Tiles**: GPSMAP and Atemos units show about **2000 in total**, including disabled maps. The page warns if you go beyond: in that case remove municipalities or delete old maps from the GPS.
2. **Internal memory**: one file per area (`<Municipality>+.img`); several can be kept. **Memory card**: the GPS reads `gmapsupp.img` (and `gmapsup1.img`... on recent models); the page picks a free name by itself.
3. **Browser memory › Free space**: deletes the municipal maps the browser keeps to avoid downloading them again (asks for confirmation; files in Downloads and on the GPS are not touched).
4. The page has no accounts and no cookies: it reads only the public files of the Drive and counts visits and loads anonymously (GoatCounter). The maps are the same as in the shared folder: the sources, licences and **disclaimer** of the PDF `Istruzioni Mappe CTRN5000 Garmin` apply.

INSTRUCTIONS FOR USE – CTRN 1:5000 MAPS IN .IMG FORMAT FOR GARMIN

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a) How the folders are arranged

1. **Folders** `(zone)_(region)__Mappe_Provinciali` – for example `_____NE_RVEN__Mappe_Provinciali`: they contain the maps of **all the provinces** of the region, one file each (`ProvVicenza.img`, `ProvVerona.img` ...). In QMapShack, pointing to this folder gives the whole region.
2. **Folders** `(zone)_(region)_Mappe_Prov(province)_Comuni` – for example `_____NE_RVEN_Mappe_ProvVicenza_Comuni`: they contain the maps of the **single municipalities** of the province.
3. Zones, in order: `_____NE` north-east, `_____NO` north-west, `___CE` centre, `__SU` south, `_IS` islands; in each region the provincial folder comes first, followed by the municipal folders, one per province. Regions: `R` + code (for example `RFVG`, `RVEN`); autonomous provinces `P` + code (`PATN` Trento, `PABZ` Bolzano). Foreign countries: `_____CH_SVIZ` Switzerland (cantons), `_____SI_SLOV` Slovenia (statistical regions), `_____AT_AUST` Austria (Länder), `_____FR_FRAN` France (border départements), instead of provinces. These instructions are in the `_____ISTRUZIONI_DISCLAIMER` folder, always at the top of the list.

Note: you will almost always use the provincial maps (point 1).

b) Installing on the Garmin GPS

With the Carica_Mappe program (the easiest way):

1. The shared folder contains `_____Carica_Mappe.dmg` (Mac with Apple chip) and `_____Carica_Mappe.exe` (Windows). Mac: double-click the dmg, drag the app into Applications, the first time open it with right-click › Open. Windows: double-click the exe; if SmartScreen appears: More info › Run anyway. If a newer version is on the Drive, the program says so at start-up.
2. **Data**: shared Drive (the link is already filled in: it downloads only the chosen maps and keeps them on the computer) or a folder on the computer with a copy of Cartografia_cond.
3. Connect the GPS (or the memory card with a reader): it appears by itself, with the maps it holds and the tiles used; maps no longer needed can be deleted from there.
4. Choose region and province: without a municipality the provincial map is loaded. Choosing a municipality: **Limitrofi** (neighbours, also across borders) or **Rettangolo** (km to North, South, East, West); the chosen municipalities are merged into one file.
5. **Carica sul GPS** (load to GPS): into the internal memory, or onto the memory card as `gmapsupp.img`. Then enable the new map on the GPS (Setup › Map › Map Select, as below).
6. **GPX tracks** (assigned areas, tracks to hand back): **Seleziona GPX…** (select GPX) opens the file chooser; several files can be taken and the step repeated from other folders, and they are added to the list (which empties when the program closes). With the box **carica anche sul GPS insieme alla mappa** (also load to the GPS with the map) the GPX of the list go into the `Garmin/GPX` folder of the GPS together with the map, replacing those with the same name.

By hand, without the program:

1. Remove any additional memory card (TrekMap or others) from the GPS. From memory cards the GPS can read only a map named `gmapsupp.img`.
2. In the internal memory of the GPS, delete any `gmapsupp.img` file.
3. Copy the file you need (province or municipality) from the Drive into the `Garmin` folder of the GPS, where `gmapsupp.img` was.

4. Switch on the GPS › Setup › Map › Map Information / Select Map (names vary slightly by model): disable all maps other than the one loaded (for example Worldwide DEM, Recreational Map of Europe) and enable the one loaded (for example ProvPordenone).
5. Setup › Display › Color Mode: choose **Day** (not Auto or Night). The maps are designed for a light background: in night mode the GPS darkens the background and contours, roads and trenches become hard to read. At night, simply lower the backlight.
6. Enjoy the maps. Suggestions and reports are welcome: claude.monbaron59@gmail.com

c) Viewing the maps on a computer

1. **From the Carica_Mappe program**: **Apri in QMapShack** (open in QMapShack) opens QMapShack with the maps of the current choice (province, municipality with neighbours or rectangle) already enabled and the view centred on the area; if the GPX list is not empty it asks whether to load those too. Each time it starts from scratch: to keep your work, before closing QMapShack right-click the project › Save As (`.qms` or `.gpx`). If QMapShack is not found, the program lets you show its path or download it.
2. **QMapShack**, free for Windows, Mac and Linux: https://github.com/Maproom/qmapshack/wiki/DocGetQMapShack
3. File › Setup Map Paths: add the folder containing the `.img` files (for example the `..._Mappe_Provinciali` folder of a region). Then, in the Maps window, right-click each map › Activate.
4. QMapShack shows several maps together and loads GPX tracks. Garmin BaseCamp shows only one map at a time.

d) Authenticity of the files

Each map shows the author and sources in the map information on the GPS. The author keeps the digital fingerprint (SHA-256) of every file distributed: if in doubt about the authenticity of a map, write to claude.monbaron59@gmail.com.

e) Technical limits

Garmin GPSMAP units have a physical limit of about 2000 tiles, including those of disabled maps. Beyond this limit the GPS disables the map that exceeds it, even though the file is still visible in the Garmin folder. To avoid this, do not load more than 1, at most 2 provinces. If a problem appears, delete the .img files that are not needed.

f) Disclaimer

These maps are a simplification needed to use them on Garmin GPS units. They are vector maps, therefore precise, built directly from the data of the regional technical maps (CTR and CTRN) and other official databases published by the individual Regions, downloaded between August and October 2026; the data date from 1979 to 2025. Sources: technical maps and databases of the Italian Regions and autonomous Provinces, of Switzerland, Slovenia, Austria and France, ISTAT and OpenStreetMap (full list in the Italian text, point g). Use of these maps is free of charge. The undersigned makes them available to all civil protection operators, volunteers or not, and declines any responsibility for inaccuracies and for improper use of these data. **Warning:** the Abruzzo data are licensed CC BY-NC: maps and files containing them may not be used for commercial purposes.

g) Borders between regions and countries

Each region and autonomous province produces its own technical map with different specifications, coding, reference systems and survey dates. All maps have been brought into the same system (WGS84) and the same symbols, but small discrepancies may remain along regional and national borders: strips of a few tens of metres without data or with overlapping data, contours, roads and rivers that do not join exactly. These are

not projection errors: each authority maps up to its own border, and the borders digitised by different administrations do not coincide perfectly.

g) Fonti dei dati e licenze

Le mappe sono elaborazioni (semplificazione, traduzione dei codici, simbologia) dei dati qui sotto. Le licenze dei dati originali restano quelle dei rispettivi enti; dove indicata, la licenza CC BY richiede di citare la fonte, come si fa qui.

1. **Friuli Venezia Giulia** – Carta Tecnica Regionale Numerica CTRN 1:5000 e Carta dei tipi forestali (IODL 2.0), Regione Autonoma Friuli Venezia Giulia (IRDAT / EagleFVG).
2. **Veneto** – Carta Tecnica Regionale 1:5000 e modello digitale del terreno DTM 5 m, Regione del Veneto (geoportale IDT).
3. **Trentino** – Carta Tecnica Provinciale, Provincia autonoma di Trento; sentieri e ghiacciai SAT.
4. **Alto Adige** – geodati della Provincia autonoma di Bolzano (carta tecnica, uso reale del suolo, sentieri, quote); curve di livello ricavate dal modello digitale del terreno 2,5 m della Provincia.
5. **Lombardia** – Database Geotopografico DBGT e Carta Tecnica CT10, isoipse dal DTM, Regione Lombardia (Geoportale, CC BY 4.0).
6. **Piemonte** – Base Dati Territoriale di Riferimento degli Enti BDTRE, Regione Piemonte (Geoportale, CC BY 4.0).
7. **Valle d'Aosta** – Carta Tecnica Regionale 1:5000 (CC0); Repertorio cartografico SCT della Regione Autonoma Valle d'Aosta (CC BY 4.0): toponimi, laghi, ghiacciai e catasto ghiacciai, confini, edifici e viabilità da deep learning 2022, rete sentieristica; Corine Land Cover di ARPA Valle d'Aosta; curve di livello ricavate dal DTM 2005/2008 a 2 m della Regione.
8. **Liguria** – Carta Tecnica Regionale 1:5000 / DB Topografico e Rete Escursionistica Ligure (REL), Regione Liguria.
9. **Emilia-Romagna** – Database Topografico Regionale DBTR (classi a libero accesso), DTM 5x5 m, uso del suolo 2023, Rete escursionistica REER, Regione Emilia-Romagna (CC BY 4.0).
10. **Toscana** – Carta Tecnica Regionale 1:10.000 / DB Topografico, Regione Toscana (Geoscopio, CC BY 4.0); isole minori (Gorgona, Capraia, Montecristo, Giglio, Giannutri) da OpenStreetMap con curve di livello ricavate da TINITALY 1.1 (INGV, CC BY 4.0).
11. **Marche** – Database Geotopografico DBGT 2019, Regione Marche (CC BY 4.0).
12. **Umbria** – Carta Tecnica Regionale numerica 1:10.000 e 1:5000, Regione Umbria (CC BY).
13. **Lazio** – Database Geotopografico DBGT 1:5000 edizione 2014, Regione Lazio (CC BY 4.0).
14. **Campania** – Carta Tecnica Numerica e Data Base Topografico 1:5000 aggiornati al 2020, Regione Campania (CC BY 4.0).
15. **Abruzzo** – Database Territoriale Regionale D.B.T.R. 1:5000 edizione 2007, Regione Abruzzo (CC BY-NC: **solo uso non commerciale**).
16. **Puglia** – Carta Tecnica Regionale 1:5000 (volo 2006), Regione Puglia, SIT Puglia (IODL 2.0).
17. **Svizzera** – swissTLM3D, swissNAMES3D, swissBOUNDARIES3D e modello del terreno swissALTI3D (curve di livello), Ufficio federale di topografia swisstopo (dati aperti, © swisstopo).
18. **Slovenia** – modello topografico DTM 1:5000, nomi geografici REZI, registro delle unità territoriali RPE e confine di Stato, Geodetska uprava Republike Slovenije GURS (CC BY 4.0).
19. **Austria** – Digitales Landschaftsmodell DLM, curve di livello DGM Höhenschichtlinien e confini dei comuni VGD, Bundesamt für Eich- und Vermessungswesen BEV (CC BY 4.0); bosco, rocce, prati e ghiacciai da OpenStreetMap.
20. **Francia** (dipartimenti di confine: Ain, Alpes-de-Haute-Provence, Hautes-Alpes, Alpes-Maritimes, Doubs, Jura, Haut-Rhin, Savoie, Haute-Savoie, Territoire de Belfort) – BD TOPO e RGE ALTI 5 m (curve di livello e quote), Institut national de l'information géographique et forestière IGN (Licence Ouverte Etalab 2.0); rocce, ghiaioni e prati da OpenStreetMap.
21. **Molise** – da OpenStreetMap (ODbL), con curve di livello ricavate da TINITALY 1.1 (INGV, CC BY 4.0), in attesa dei dati regionali.
22. **Basilicata** – Database Geotopografico DBGT 2013, Regione Basilicata, RSDI (CC BY 4.0).

23. **Calabria** – Database Geotopografico DBGT 1:5000 2007–2008, Regione Calabria (dati aperti).
24. **Sicilia** – Carta Tecnica Regionale 1:10.000 2012–2013, Regione Siciliana, SITR (CC BY 4.0); boschi integrati da OpenStreetMap (ODbL).
25. **Sardegna** – Database Geotopografico 1:10.000 (DBGT_10K_22), Regione Autonoma della Sardegna.
26. **Confini e nomi dei comuni** – ISTAT, Limiti amministrativi al 1° gennaio 2026 (CC BY).
27. **OpenStreetMap** – © OpenStreetMap contributors, licenza ODbL (openstreetmap.org/copyright): mare e lagune (osmdata.openstreetmap.de), laghi del Friuli, laghi e canali del Veneto, piste da sci, strade principali di Trentino e Alto Adige, copertura del suolo dell'Austria, parchi di Friuli, Valle d'Aosta e Trentino.

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